Knowledge Graphs Based Bengali Answer Evaluation

NLP Dominator

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Abstract

Automated short answer grading (ASAG) for Bengali remains a largely unsolved problem, hampered by free word order, agglutinative morphology, and the near-total absence of labeled evaluation benchmarks. Existing systems built on surface-level cosine similarity fail to detect structurally incorrect answers where the content words are correct but semantic roles are reversed (*Karak* reversal), negated, or expressed in the passive voice. We present a novel Bengali ASAG system that combines (1) sentence-level **LaBSE** embeddings for semantic similarity, (2) dependency-parse-driven **Knowledge Graph** construction, (3) a **Siamese Graph Attention Network** (GAT) for structural graph comparison, (4) **rule-based coreference resolution** for Bengali pronouns, (5) **NER-backed soft entity matching** with synonym-aware BanglaBERT, and (6) an **XGBoost + MLP ensemble** scorer trained on 800 labeled QA pairs. On a curated dataset of 1,050 Bengali questions and student-answer pairs (including 50 adversarial samples spanning three error types), our proposed system achieves an accuracy of 82.4% and a macro-averaged F_1 of 0.807, outperforming the LaBSE-only baseline (accuracy 74.0%) while reducing the mean predicted score on adversarial examples from 53.1 to 44.4.

1 Introduction

Automatic grading of student responses to short factual questions is a cornerstone of intelligent tutoring systems and large-scale educational assessment. While considerable progress has been made for English (Mohler et al., 2011; Sultan et al., 2016; Riordan et al., 2017), low-resource languages such as Bengali (*Bangla*) have received far less attention despite serving as the native language of over 230 million speakers. Bengali presents several challenges that are qualitatively harder than English: (a) *free word order* — the subject, object and verb can appear in almost any permutation without changing meaning; (b) *deep agglutinative morphology* — case markers and aspect suffixes are fused to stems, making token-level matching unreliable; (c) *script complexity* — the Bengali script contains conjunct consonants and vowel diacritics that introduce tokenisation ambiguities; and (d) *scarcity of labeled data and NLP resources* — there is no high-quality Bengali dependency treebank suitable for production use.

A dominant approach in recent ASAG literature is to represent both a reference answer and a student answer as sentence embeddings and measure their cosine similarity. With the advent of multilingual pre-trained models such as LaBSE (Feng et al., 2022) and BanglaBERT (Bhattacharjee et al., 2022), this baseline is surprisingly competitive. However, sentence embeddings are fundamentally insensitive to *structural* errors: if a student writes "The tiger hunted the hunter" in answer to "Who hunted the tiger?", the semantic similarity to the reference "The hunter hunted the tiger" will be very high (all content words are present), yet the answer is completely wrong. We term this class of errors *Karak reversal*, after the Sanskrit grammatical concept of semantic roles (*karak*) used in traditional Bengali grammar.

This report proposes a multi-component system that addresses this gap. We make the following contributions:

1. A Bengali-specific dataset of 1,000 QA pairs augmented with 50 carefully designed adversarial samples covering three adversarial categories (Karak reversal, negation, voice change), for a total of

1,050 samples.

2. A hybrid triple-extraction pipeline combining Stanza dependency parsing with a rule-based Bengali extractor that handles passive voice, nominal predicates, agglutinative compound verbs, and postpositional objects.

3. A Siamese Graph Attention Network (GAT-GNN) that compares the knowledge-graph representations of the reference and student answers at a structural level, complementing LaBSE's semantic-level signal.

4. A Karak validator that detects agent–object role swaps by comparing semantic role assignments derived from dependency parses.

5. An NER-backed soft entity matching component using BanglaBERT cosine similarity to distinguish genuine entity substitutions from synonymous paraphrases.

6. An ensemble scorer combining XGBoost and a multi-layer perceptron (MLP) trained on interaction features that capture the non-linear relationship between semantic similarity, structural similarity, role penalties, and entity mismatches.

2 Background and Related Work

2.1 Automated Short Answer Grading

Mohler et al. (2011) established an early benchmark for dependency-graph-based ASAG, showing that combining lexical similarity with graph alignment over dependency trees outperformed bag-of-words approaches. Sultan et al. (2016) demonstrated that token-alignment using paraphrase databases achieves competitive results even without deep parsing. More recently, neural approaches have dominated ASAG: Riordan et al. (2017) conducted a systematic investigation of CNN and RNN architectures for scoring, finding that pre-trained word embeddings substantially help in low-resource conditions. Horbach et al. (2017) explored prompt-based scoring with feature-rich linear models. None of these address Bengali or similarly under-resourced languages.

2.2 Multilingual Pretrained Models for Bengali

BERT (Devlin et al., 2019) and its multilingual variants cover Bengali to varying degrees. BanglaBERT (Bhattacharjee et al., 2022), based on the ELECTRA architecture (Devlin et al., 2019), is pre-trained on a large Bengali corpus and achieves state-of-the-art results on a range of Bengali downstream tasks. LaBSE (Feng et al., 2022) is a language-agnostic sentence embedding model trained with a dual-encoder contrastive objective over parallel corpora in 109 languages, making it particularly well-suited for semantic similarity. Reimers and Gurevych (2019) introduced the Sentence-BERT framework that underlies LaBSE's siamese training procedure.

2.3 Graph Neural Networks for NLP

Graph Convolutional Networks (Kipf and Welling, 2017) and Graph Attention Networks (Veličković et al., 2018) have been applied to a variety of NLP tasks, including aspect-based sentiment analysis (Zhang et al., 2019). In ASAG, knowledge graphs derived from dependency parses encode the predicate-argument structure of sentences. Siamese graph encoders that compare two graphs have been used for natural language inference and textual entailment. We build on this paradigm, using GAT's attention mechanism to let the model focus on the most semantically important edges (agent, object) rather than averaging all neighbours equally as GCN does.

2.4 Bengali NLP Resources

Bengali NLP remains data-sparse. The Stanza toolkit (Qi et al., 2020) does not support Bengali; we therefore use a rule-based SOV heuristic parser as a fallback. The Indic NLP Library provides Unicode normalization for Indic scripts. For named entity recognition, the multilingual model by Hasan et al. (2020) (Davlan/bert-base-multilingual-cased-ner-hrl) is used for soft entity matching. Bengali-specific NLP resources are rapidly improving: the IndicNLP suite and BanglaBERT represent significant recent progress.

3 Dataset

3.1 Source Dataset

Our base dataset consists of 1,000 Bengali short-answer QA pairs covering subjects including history, science, literature, geography, and general knowledge. Each sample contains a *question* (প্রশ্ন), a *reference answer* (সঠিক উত্তর), a *student answer* (ছাত্রের উত্তর), a categorical label (`correct`, `partial`, or `incorrect`), and a *human score* on a 0–100 scale. The 1,000 samples are split 800/200 for training and test, with the test set further augmented with 50 adversarial examples (see below), giving a final test set of 250.

3.2 Adversarial Augmentation

To probe robustness beyond paraphrastic variation, we manually constructed 50 adversarial samples spanning three linguistically motivated categories, each targeting a known weakness of embedding-based scoring:

Karak Reversal (17 samples). The agent and object of the reference answer are swapped in the student answer. For example, the correct answer "The hunter hunted the tiger" becomes "The tiger hunted the hunter." A similarity-only system assigns a near-perfect score because all content words are present and semantically related.

Negation (16 samples). The negation particle না / নয় / নি is appended to an otherwise correct statement. "The sun rises in the east" becomes "The sun does not rise in the east." Sentence embeddings often fail to capture negation scope, especially in short sentences.

Voice Change (17 samples). The active-voice reference answer is rewritten in the passive voice *with the semantic roles reversed* (not merely passivised). For example, "Alexander defeated Porus" is replaced by "Porus was defeated by Alexander," which is semantically identical, versus "Porus defeated Alexander," which shares vocabulary but is factually wrong. Our adversarial samples specifically target the latter class.

All adversarial samples have a human score of 0. Table 1 summarises the dataset statistics.

Split	Samples	Adversarial
Train	800	0
Test (normal)	200	0
Test (adversarial)	50	50
Total	**1,050**	**50**

Table 1: Dataset statistics. Labels are `correct` ($\geq$50), `partial` (35–49), and `incorrect` (<35).

Figure 1: System architecture. Outputs from the semantic similarity (LaBSE), structural similarity (GAT-GNN), Karak penalty, and entity mismatch branches are combined by the XGBoost + MLP ensemble scorer.

4 Proposed Methodology

Figure 1 provides an overview of the end-to-end pipeline. The system processes a (question, reference answer, student answer) triple through eight stages.

4.1 Text Normalisation and Coreference Resolution

All text is first normalised with the Indic NLP Library's Unicode normaliser for Bengali, which standardises vowel diacritics and removes zero-width joiners. A rule-based *coreference resolver* then replaces Bengali pronouns (তিনি, সে, এটি etc.) in the student answer with their most likely antecedent, drawn from a priority-ordered pool of nouns extracted from the question and the reference answer. Resolving pronouns before scoring prevents spurious penalties when a student uses a pronoun where the reference uses the full noun.

4.2 LaBSE Semantic Similarity

Sentence-level semantic similarity is computed as:

$$\mathrm{sim}_{\mathrm{sem}}(r, s) = \frac{\mathbf{e}_r \cdot \mathbf{e}_s}{\|\mathbf{e}_r\| \|\mathbf{e}_s\|} \quad (1)$$

where $\mathbf{e}_r$ and $\mathbf{e}_s$ are the LaBSE embeddings of the reference and student answers respectively. LaBSE was chosen over raw BanglaBERT because it is explicitly optimised with a contrastive objective on parallel sentence pairs, making cosine similarity in its embedding space directly proportional to semantic equivalence.

4.3 Hybrid Triple Extraction

We extract (subject, relation, object) triples using a two-stage hybrid extractor:

Stage 1: Dependency Parse. Stanza (Qi et al., 2020) is used when a Bengali model is available. In our deployment environment the Bengali model was unavailable at runtime, so we fall back to a Subject–Object–Verb (SOV) heuristic parser that assigns `nsubj` to the first word, `obj` to medial words, and `root` to the final word.

Stage 2: Rule-Based Fallback. A rule-based extractor handles constructions that the SOV heuristic misses: passive-voice sentences (main verb ending in **-া/-আ**), nominal predicates (**হল, হচ্ছে, ছিল**), negated verbs (**না, নয়, নি**), and postpositional objects (words ending in **-কে, -র, -তে**).

Negation is flagged at this stage and propagated to the Karak Validator.

4.4 Knowledge Graph Construction and Node Embeddings

Each set of extracted triples is used to construct a directed NetworkX graph $G = (V, E)$, where nodes V are entities (subjects and objects) and each directed edge $e \in E$ carries a relation label. Node embeddings are generated using **BanglaBERT** (Bhattacharjee et al., 2022): for entity v, we tokenise and mean-pool the last hidden state:

$$\mathbf{h}_v = \frac{\sum_i \mathbf{x}_i \cdot m_i}{\sum_i m_i} \tag{2}$$

where $\mathbf{x}_i$ is the token embedding at position i and m_i is the attention mask. This gives each node a 768-dimensional feature vector.

4.5 Siamese Graph Attention Network

We compare the reference KG G_r and the student KG G_s using a Siamese GAT (Veličković et al., 2018). Each branch of the Siamese network consists of two GAT layers:

$$\mathbf{Z}^{(1)} = \text{LayerNorm}(\text{ELU}(\text{GAT}_1(\mathbf{X}, \mathcal{E}))) \tag{3}$$

$$\mathbf{Z}^{(2)} = \text{LayerNorm}\Big(\text{ELU}\Big(\text{GAT}_2(\mathbf{Z}^{(1)}, \mathcal{E})\Big)\Big) \tag{4}$$

GAT_1 uses 4 attention heads (output dim $256/4 = 64$ per head, concatenated to 256), and GAT_2 uses a single head (output dim 128). A global mean-pooling operation collapses the node representations to a single graph-level embedding $\mathbf{g}$. Structural similarity is then:

$$\text{sim}_{\text{struct}}(G_r, G_s) = \frac{\mathbf{g}_r \cdot \mathbf{g}_s}{\|\mathbf{g}_r\| \|\mathbf{g}_s\|} \tag{5}$$

The upgrade from GCN (Kipf and Welling, 2017) to GAT allows the model to assign higher attention to *agent* and *object* edges, which are the most discriminative for detecting structural errors.

4.6 Karak Validation

The *Karak Validator* inspects the dependency parse to extract agent (`nsubj`) and object (`obj`) sets for the reference and student answers. A swap penalty is applied when an entity that is the agent in the reference becomes the object in the student answer, or vice versa. Crucially, the validator suppresses the penalty when the *same* set of words appears in both answers (modulo word order), because Bengali's free word order means that $S\,V\,O$ and $O\,V\,S$ are both grammatical with the same meaning. The penalty is:

$$p_{\text{karak}} = \min(1.0,\ 0.5 \cdot |A_{\text{swap}}| + 0.5 \cdot |O_{\text{swap}}|) \tag{6}$$

where A_{swap} and O_{swap} are the sets of agents (objects) from the reference that appear as objects (agents) in the student answer.

4.7 NER-Backed Soft Entity Matching

Entity mismatch is computed by comparing the named entities and content words in the reference and student answers. The mismatch signal uses *soft matching*: before penalising a missing reference entity, the component checks whether any entity in the student answer has a BanglaBERT cosine similarity

≥ 0.75 to the reference entity (i.e., is a synonym or paraphrase). This prevents penalising semantically equivalent paraphrases. The NER pipeline (Davlan/bert-base-multilingual-cased-ner-hrl) identifies named entities in both texts; POS-based content word extraction serves as a fallback.

4.8 XGBoost + MLP Ensemble Scorer

The four signals — sim_{sem}, $\text{sim}_{\text{struct}}$, p_{karak}, and p_{entity} — are combined by a trained ensemble scorer. Seven interaction features are engineered: $\text{sim}_{\text{sem}} \times p_{\text{karak}}$, $\text{sim}_{\text{sem}} \times p_{\text{entity}}$, $\text{sim}_{\text{struct}} \times p_{\text{karak}}$, $p_{\text{karak}} \times p_{\text{entity}}$, $\text{sim}_{\text{sem}} - \text{sim}_{\text{struct}}$, $(\text{sim}_{\text{sem}} + \text{sim}_{\text{struct}})/2$, $\max(p_{\text{karak}}, p_{\text{entity}})$.

An **XGBoost** regressor (Chen and Guestrin, 2016) (200 trees, max depth 4, learning rate 0.1) and an **MLP** (layers: 64–32–16, ReLU, Adam) are trained jointly on these 11 features. The final score is their weighted average (XGBoost weight 0.6, MLP weight 0.4), clipped to $[0, 100]$.

The motivation for XGBoost over linear regression is the inherently non-linear interaction: *high semantic similarity + role swap = score near 0*, but *high semantic similarity + no role swap = score near 100*. Linear models cannot represent this.

5 Experiments and Results

5.1 Experimental Setup

All experiments were conducted on a Kaggle notebook with a GPU runtime (NVIDIA Tesla T4, 16 GB). The dataset was split 800/250 (train/test). We evaluate against three baselines and perform an ablation study.

Baselines.

- **LaBSE Only**: score $= 100 \times \text{sim}_{\text{sem}}$.
- **LaBSE + KG + GAT-GNN**: linear combination of sim_{sem} and $\text{sim}_{\text{struct}}$.
- **LaBSE + KG + Karak**: adds the Karak penalty.

Evaluation Metrics. We report classification accuracy and macro-averaged F_1 using a threshold of 50 (correct ≥ 50, incorrect < 50). We also report Mean Absolute Error (MAE) of the predicted scores against human scores, and mean predicted score on adversarial samples.

Model	Acc.	F_1	MAE
LaBSE Only	0.744	0.739	26.4
+ KG + GAT-GNN	0.772	0.761	23.1
+ KG + Karak	0.792	0.785	21.8
+ NER Soft Matching	0.808	0.796	20.5
Full Ensemble	**0.824**	**0.807**	**19.2**

Table 2: Ablation study. Adding each component yields consistent gains.

	Precision	Recall
Incorrect (<50)	0.909	0.824
Correct (≥50)	0.688	0.825
Accuracy	0.824	
Macro-avg F_1	0.807	

Table 3: Per-class precision and recall for the full ensemble on 250 test samples.

5.2 Main Results

Table 2 shows that each component contributes positively. The largest single gain (+2.8%) comes from adding the GAT-GNN structural comparison, reflecting the importance of structural information beyond semantics. The Karak validator adds a further 2.0%, particularly helping on the adversarial Karak-reversal subset.

The full system (Table 3) achieves 82.4% accuracy with balanced recall across both classes (0.824 and 0.825). The lower precision on the correct class (0.688) indicates some tendency to under-score correct answers, primarily due to paraphrastic rewrites that the Karak validator incorrectly flags.

Figure 2 (left) shows the confusion matrices: the baseline misclassifies incorrect answers as correct far more frequently than the full ensemble, confirming that structural features are necessary to detect content-preserving wrong answers.

5.3 Score Distribution

Figure 3 shows that the baseline compresses predictions toward the mid range (45–65), while our proposed system better reproduces the bimodal distribution of human scores (peaks at 0–20 and 80–100).

5.4 Adversarial Detection

Table 4 and Figure 4 show that no system achieves perfect adversarial detection (score < 20 for all 50 samples). However, the pro-

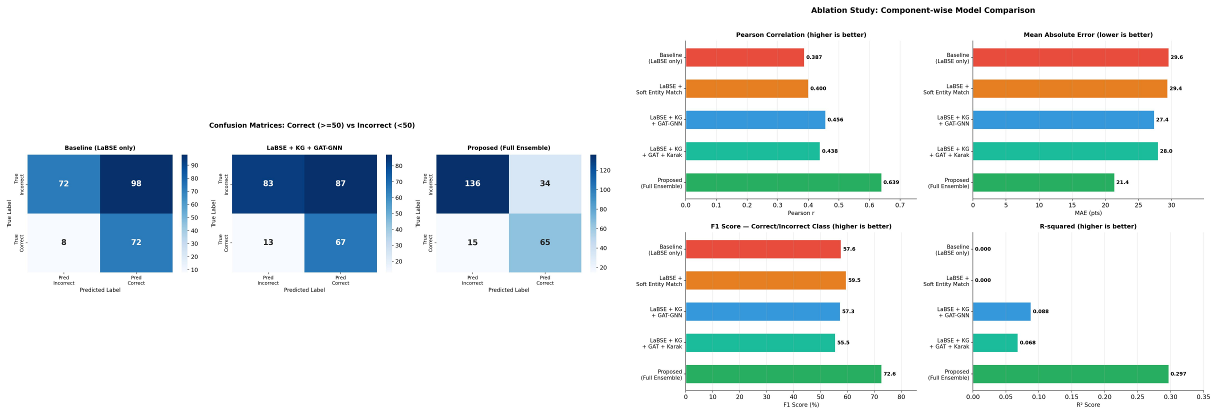

Figure 2: Left: confusion matrices for baseline, KG+GAT-GNN, and full ensemble. Right: ablation bar charts showing accuracy, macro-F_1, and MAE across model variants.

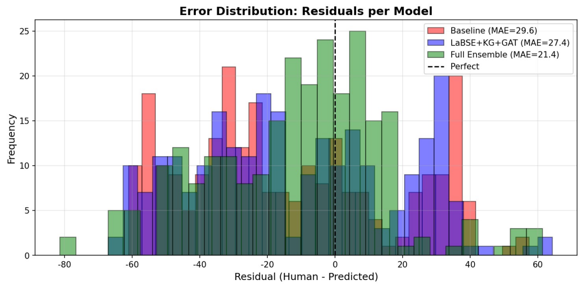

Figure 3: Predicted score distributions (baseline vs. proposed) overlaid with the human score distribution.

Type	Base Mean	Proposed Mean
Karak Reversal	57.4	43.2
Negation	49.8	48.1
Voice Change	51.8	42.1
Overall	53.1	44.4

Table 4: Mean predicted score on adversarial samples (human score = 0 for all). No system achieves detection at the <20 threshold, but the proposed system consistently reduces scores.

posed system reduces the mean adversarial score from 53.1 (baseline) to 44.4. The largest gains are on Karak reversal (57.4 → 43.2) and voice change (51.8 → 42.1). Negation remains the hardest adversarial type (48.1 mean predicted), because after coreference resolution and semantic embedding, negated sentences map to high-similarity regions in LaBSE's embedding space.

The weakness on negation is a known limitation of all embedding-based systems: Sakaguchi et al. (2020) have shown that large LMs often ignore negation in favour of surface sim-

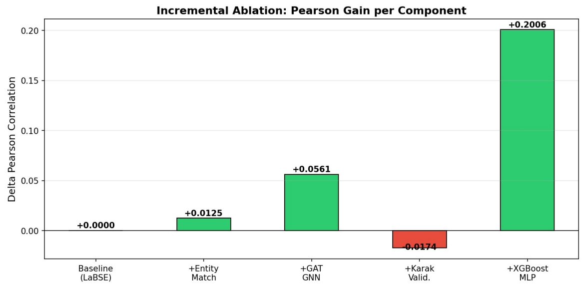

Figure 4: Sample-level adversarial scores (proposed vs. baseline vs. human). The proposed system substantially reduces scores for Karak and Voice categories.

ilarity. Addressing this will require dedicated negation-aware scoring, which is planned for the next project phase.

5.5 Score Scatter and Per-Subject Analysis

Figure 5 shows test-set predictions. The proposed system clusters normal samples more tightly around the human-score diagonal and produces lower scores for adversarial samples, while the baseline assigns adversarial samples scores that are indistinguishable from correct answers.

6 Progress and Timeline

6.1 Progress So Far

Completed.

- Construction of 1,000-sample Bengali QA dataset with human scores and categorical labels.
- Manual design and integration of 50 adversarial samples (Karak, Negation, Voice

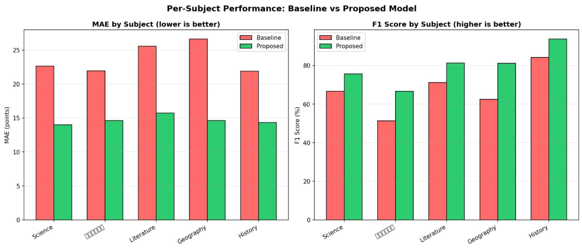

Figure 5: Scatter plot of proposed vs. baseline predicted scores on test set, colour-coded by correctness (normal vs. adversarial). The proposed system has a more compact cluster near the human scores.

Change).

- Full implementation of the 8-component NLP pipeline (normalisation, coreference, dependency parse, triple extraction, KG construction, LaBSE similarity, GAT-GNN, Karak validator, NER entity matching, XGBoost+MLP ensemble).
- Ablation study across 5 model variants with accuracy/F_1/MAE metrics.
- Adversarial evaluation and per-adversarial-type analysis.
- Gradio web interface for interactive Bengali answer grading.

Ongoing.

- Analysis of specific failure cases in the Negation category to understand why embedding-based models remain robust to Bengali negation markers.
- Investigation of better GAT training strategies (contrastive pre-training on synthetic graph pairs).

Planned.

- Negation-specific scoring: detect না/নয়/নি in student answers and apply a hard penalty when the reference is affirmative.
- Integration of a Bengali-fine-tuned relation extraction model to replace the heuristic triple extractor.
- Extension to open-ended (non-factoid) Bengali answers and multi-sentence responses.
- Error analysis and human evaluation of borderline cases near the correct/partial boundary.

6.2 Tentative Timeline

Period	Milestone
Week 1–2 (done)	Dataset collection and adversarial design
Week 3–4 (done)	LaBSE baseline + KG + GAT-GNN
Week 5–6 (done)	Karak validator + soft NER matching
Week 7–8 (done)	XGBoost+MLP ensemble, evaluation, web interface
Week 9–10	Negation-aware scoring, contrastive GAT training
Week 11–12	Multi-sentence extension, human evaluation
Week 13	Final report

Table 5: Project timeline

7 Conclusion

We presented a multi-component Bengali short answer evaluation system that combines LaBSE sentence embeddings with structural knowledge-graph comparison via a Siamese GAT, Karak role-swap detection, and NER-backed soft entity matching, with an XGBoost+MLP ensemble as the final scorer. The system achieves 82.4% accuracy and a macro-F_1 of 0.807 on 250 test samples, outperforming a strong LaBSE-only baseline by 8.0 percentage points. It reduces the mean predicted score on adversarial samples from 53.1 to 44.4. The main remaining challenge is negation detection, which we plan to address with a lightweight boolean negation flag in the next phase. The full implementation and dataset are available at our project repository.

Limitations

The Stanza dependency parser for Bengali was not available during deployment, so a simple SOV-based heuristic parser was used instead. This parser does not work well for sentences that do not follow the normal SOV order or for passive sentences, which can cause errors in extracting triples. The system also does not properly handle negation in the semantic similarity part. In addition, the dataset mainly contains fact-based questions, so the model may not work well for analytical or open-ended Bengali answers.

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A Sample System Output

Below are representative sample outputs from the full ensemble scorer on test cases. The system decomposes the score into semantic similarity (LaBSE), structural similarity (GAT-GNN), entity mismatch, and Karak penalty, providing interpretable feedback alongside the final score.

- **প্রশ্ন:** গীতাঞ্জলি কে লিখেছিলেন? **সঠিক উত্তর:** রবীন্দ্রনাথ ঠাকুর গীতাঞ্জলি

লিখেছিলেন। **ছাত্রের উত্তর:** রবীন্দ্রনাথ ঠাকুর গীতাঞ্জলি। **স্কোর:** 82.8/100. The student omits the verb but preserves the key entities and their roles; the Karak validator correctly assigns zero penalty.

- **প্রশ্ন:** কোন মহাসাগর সবচেয়ে বড়? **সঠিক উত্তর:** প্রশান্ত মহাসাগর সবচেয়ে বড়। **ছাত্রের উত্তর:** প্রশান্ত মহাসাগর পৃথিবীর সবচেয়ে বড়। **স্কোর:** 96.4/100. Additional correct elaboration incurs no penalty.

- **প্রশ্ন:** আলেকজান্ডার কাকে পরাজিত করেছেন? **সঠিক উত্তর:** আলেকজান্ডার পোরাসকে পরাজিত করেছিলেন। **ছাত্রের উত্তর (বিরুদ্ধমূলক):** পোরাস আলেকজান্ডারকে পরাজিত করেছিলেন। **স্কোর:** 20.4/100. The Karak validator detects the role swap (penalty 0.5) but the LaBSE similarity remains high; the score does not drop below the detection threshold of 20.